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Rh-Catalyzed oxidative homo-coupling cyclization of 2,3-allenols to conjugated furylenones†

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An unexpected rhodium-catalyzed highly regio- and stereo-selective bimolecular nucleophilic oxyrhodation/insertion/1,4-Rh migration/β-H elimination reaction of 2,3-allenols affording (2,5-dihydrofuran-3-yl) but-2(*E*)-en-1-one derivatives is reported. Optically active products may be prepared from optically active 2,3-allenols without obvious racemization. The synthetic potentials have been demonstrated.

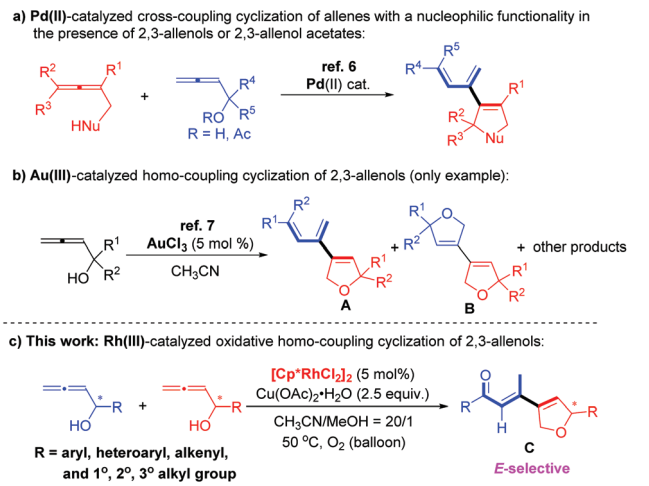
Introduction

Cyclization involving two allenes has become a very important approach for the synthesis of carbo- and hetero-cyclic compounds.[1,2] In this area, Pd(II)-catalyzed cross-coupling cyclization of allenes with a nucleophilic functionality in the presence of 2,3-allenols or 2,3-allenyl acetates leading to highly versatile 2,5-dihydrofuran derivatives[3–5] with a synthetically useful conjugated 1,3-diene moiety[6] has been well established (Scheme 1a). Besides the palladium catalysis, Hashmi and co-workers also realized the cyclization of tertiary 2,3-allenols under the catalysis of $AuCl_3$, affording a mixture of **A**-type monocyclic products and **B**-type double cyclizative coupling products together with other by-products (Scheme 1b).[7] In all these cases, a β-O elimination process occurred to generate conjugated dienes. Here, we wish to report a Rh(III)-catalyzed regio- and stereo-selective bimolecular cyclization of 2,3-allenols, in which one allenol cyclized to form the dihydrofuran ring while the second molecule of allenol provided the conjugated enone unit, affording **C**-type derivatives with an exclusive *E*-selectivity (Scheme 1c).

Results and discussion

Initially, the reaction conditions were optimized using 2,3-allenol **1a** (0.3 mmol) in the presence of $[Cp^*RhCl_2]_2$ (2.5 mol%) in methanol under an air atmosphere. The expected products, such as the homo-coupling cyclization products **A1** and **B1** or the cycloisomerization product **D1**, were not detected, and we merely observed the decomposition of 2,3-allenol **1a** (Table 1, entry 1). When $Cu(OAc)_2{\cdot}H_2O$ (2.0 equiv.) was added, the reaction afforded a new product, which was later identified as enone (*E*)-**2a**, in 15% NMR yield after 36 hours, unexpectedly (Table 1, entry 2). After a series of solvents were screened, it was found that the reaction in CH_3CN gave (*E*)-**2a** in a slightly better yield of 20% with 23% recovery of **1a** (Table 1, entries 3–8).

Considering the possible regeneration of the catalytically active metal catalyst, the reaction was conducted under a pure O_2 atmosphere and a higher yield of (*E*)-**2a** was observed (Table 2, entry 1). Increasing the loading of the Rh-catalyst to 4 mol% led to 38% yield of this product (Table 2, entry 2). Further investigation of some mixed solvents showed that the reaction in CH_3CN/MeOH (20/1) gave the best results (Table 2, entries 3–6). Changing the ratio of CH_3CN/MeOH all failed to improve the yield (Table 2, entries 7 and 8). Further increasing



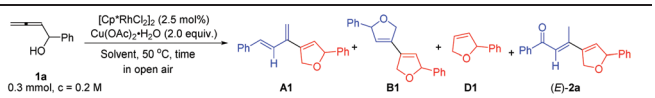
Scheme 1 Transition metal-catalyzed cross-coupling cyclization of allenes in the presence of allenols or their derivatives.

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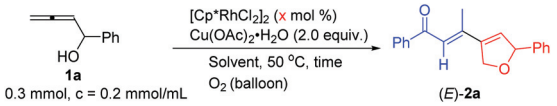
Table 1 Solvent effect[a]



Entry	Solvent	Time (h)	Yield of (*E*)-**2a**[b] (%)	Recovery of **1a**[b] (%)
1[c]	MeOH	14	0	0
2	MeOH	36	15	0
3	DCE	72	15	Trace
4	DMF	72	0	33
5	DMA	72	0	25
6	DMSO	72	2	53
7	Dioxane	72	7	18
8	CH_3CN	72	20	23

[a] Reaction conditions: **1a** (0.3 mmol), $[Cp^*RhCl_2]_2$ (2.5 mol%), and Cu $(OAc)_2{\cdot}H_2O$ (0.6 mmol) in the solvent (1.5 mL) at 50 °C under an air atmosphere (balloon) unless otherwise noted. [b] Determined by the ^1H NMR analysis of the crude products with mesitylene as the internal standard. [c] Without $Cu(OAc)_2{\cdot}H_2O$.

Table 2 Optimization of the reaction conditions under an O_2 atmosphere[a]



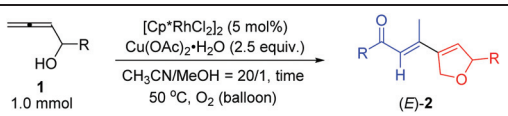
Entry	*x*	Solvent	Time (h)	Yield/recovery (*E*)-**2a**/**1a**[b] (%)
1	2.5	CH_3CN	60	30/9
2	4	CH_3CN	43	38/4
3	4	CH_3CN/H_2O (20/1)	36	20/3
4	4	CH_3CN/HOAc (20/1)	36	40/0
5	4	CH_3CN/NEt_3 (20/1)	36	0/22
6	4	CH_3CN/MeOH (20/1)	36	47/8
7	4	CH_3CN/MeOH (10/1)	41	40/14
8	4	CH_3CN/MeOH (50/1)	41	42/9
9	5	CH_3CN/MeOH (20/1)	43	49/4
10	6	CH_3CN/MeOH (20/1)	43	47/0
11	7	CH_3CN/MeOH (20/1)	43	38/0
12[c]	5	CH_3CN/MeOH (20/1)	48	43/3
13[d]	5	CH_3CN/MeOH (20/1)	38	49/0
14[d,e]	5	CH_3CN/MeOH (20/1)	60	48 (46)[f]/2
15[d,e,g]	5	CH_3CN/MeOH (20/1)	38	58 (58)[f]/0
16[d,e,h]	5	CH_3CN/MeOH (20/1)	12.5	62 (62)[f]/0
17[d,e,i]	5	CH_3CN/MeOH (20/1)	8	60/0

[a] Reaction conditions: **1a** (0.3 mmol), $[Cp^*RhCl_2]_2$ (*x* mol%), and Cu $(OAc)_2{\cdot}H_2O$ (0.6 mmol) in the solvent (1.5 mL) at 50 °C under an O_2 atmosphere (balloon) unless otherwise noted. [b] Determined by the ^1H NMR analysis of the crude products with mesitylene as the internal standard. [c] $Cu(OAc)_2{\cdot}H_2O$ (1.5 equiv.) was added. [d] $Cu(OAc)_2{\cdot}H_2O$ (2.5 equiv.) was added. [e] 1.0 mmol scale. [f] Isolated yield. [g] 0.48 M. [h] 0.95 M. [i] 1.9 M.

the loading of the Rh-catalyst and $Cu(OAc)_2{\cdot}H_2O$ showed that 5 mol% of Rh-catalyst and 2.5 equiv. of $Cu(OAc)_2{\cdot}H_2O$ gave the best result (Table 2, entries 9–13). However, the reaction was rather sluggish when it was scaled up to 1 mmol (Table 2, entry 14). Luckily, performing the reaction at a higher concentration of **1a** remarkably improved the rate and yield, affording (*E*)-**2a** in 62% isolated yield at a concentration of 0.95 M (Table 2, entries 15–17). Thus, the conditions described in entry 16 were chosen as the optimal conditions.

Halogen atoms (F, Cl, and Br) substituted on the *para*-, *ortho*- or *meta*-position of the phenyl groups could all be tolerated, leaving opportunity for further synthetic elaboration (Table 3, entries 2–6). The reaction of allenols bearing an electron-donating group-substituted phenyl group, **1g** and **1h**, afforded the products (*E*)-**2g** and (*E*)-**2h** in 57% and 44% yields, respectively (Table 3, entries 7 and 8). In addition, 2-naphthyl substituted allenol **1i** could also be transformed into (*E*)-**2i** in 52% yield (Table 3, entry 9). Furthermore, heteroaryl groups such as 2-furyl and 2-thienyl substituted allenols also led to the formation of (*E*)-**2j** and (*E*)-**2k** in decent yields (Table 3, entries 10 and 11). Furthermore, the reaction of 1-cinnamyl-2,3-butadienol (*E*)-**1l** resulted in the formation of (*E*,*E*,*E*)-**2l** with four C=C bonds in 41% yield (Table 3, entry 12). Primary alkyl group substituted 2,3-allenols could also react smoothly to afford (*E*)-**2m**–(*E*)-**2r** in 51%–64% yields (Table 3, entries 13–18). Primary alkyl groups containing functionalities such as chlorine and a free hydroxyl group and secondary or tertiary alkyl substituted allenols also worked well (Table 3, entries 19 and 22). A gram-scale reaction could be easily realized (Table 3, entries 6 and 14).

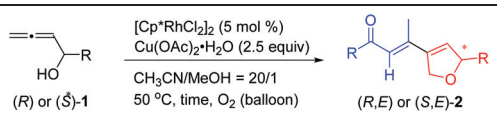
Table 3 The scope of 2,3-allenols[a]



Entry	R	Time (h)	Yield of (*E*)-**2**[b] (%)
1	Ph (**1a**)	12.5	62 ((*E*)-**2a**)
2	**4-F**C_6H_4 (**1b**)	11	57 ((*E*)-**2b**)
3	**2-Cl**C_6H_4 (**1c**)	7.5	53 ((*E*)-**2c**)
4	**3-Cl**C_6H_4 (**1d**)	10.5	54 ((*E*)-**2d**)
5	**3-Br**C_6H_4 (**1e**)	11	53 ((*E*)-**2e**)
6[c]	**4-Br**C_6H_4 (**1f**)	12	52 ((*E*)-**2f**)
7	**4-MeO**C_6H_4 (**1g**)	11	57 ((*E*)-**2g**)
8[d]	Benzodioxol-5-yl (**1h**)	12.5	44 ((*E*)-**2h**)
9	2-Naphthyl (**1i**)	14	52 ((*E*)-**2i**)
10	2-Furyl (**1j**)	12	43 ((*E*)-**2j**)
11	2-Thienyl (**1k**)	11	49 ((*E*)-**2k**)
12	Cinnamyl ((*E*)-**1l**)	13	41 ((*E*,*E*,*E*)-**2l**)
13	*n*-C_6H_{13} (**1m**)	18	64 ((*E*)-**2m**)
14[e]	*n*-C_9H_{19} (**1n**)	13	62 ((*E*)-**2n**)
15	*n*-$C_{11}H_{23}$ (**1o**)	12	63 ((*E*)-**2o**)
16	i-Bu (**1p**)	9	51 ((*E*)-**2p**)
17	Bn (**1q**)	10	54 ((*E*)-**2q**)
18	$(CH_2)_2$Ph (**1r**)	11	62 ((*E*)-**2r**)
19	$(CH_2)_5$**Cl** (**1s**)	12	62 ((*E*)-**2s**)
20	$(CH_2)_5$**OH** (**1t**)	12	56 ((*E*)-**2t**)
21	Cy (**1u**)	12.5	60 ((*E*)-**2u**)
22	*t*-Bu (**1v**)	10	50 ((*E*)-**2v**)

[a] Reaction conditions: **1** (1.0 mmol), $[Cp^*RhCl_2]_2$ (0.05 mmol), Cu $(OAc)_2{\cdot}H_2O$ (2.5 mmol), and MeOH (50 μL) in CH_3CN (1.0 mL) at 50 °C under an O_2 atmosphere (balloon) unless otherwise noted. [b] Isolated yield. [c] 10 mmol scale, 2 mol% byproduct (*E*)-**8f** was obtained, see the ESI† for details. [d] Besides (*E*)-**2h**, piperonyl aldehyde (3.1 mg, 2%) and a mixture of other byproducts (8.1 mg) were also obtained. [e] 5.82 mmol scale.

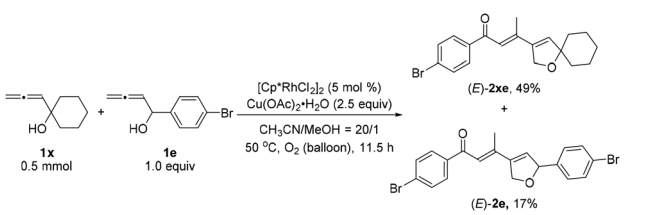
Table 4 Synthesis of optically active (2,5-dihydrofuran-4-yl)but-2-en-1-one derivatives[a,b]



Entry	(*R*) or (*S*)-**1**	Time (h)	(*R*,*E*) or (*S*,*E*)-**2**	Yield (%)	ee (%)
1	(*S*)-**1a**, 98% ee	12.5	(*S*,*E*)-**2a**	56	98
2	(*S*)-**1b**, >99% ee	12	(*S*,*E*)-**2b**	61	>99
3	(*S*)-**1e**, 95% ee	12	(*S*,*E*)-**2e**	48	93
4	(*S*)-**1j**, 96% ee	12	(*S*,*E*)-**2j**	42	95
5	(*R*)-**1m**, 98%	12	(*R*,*E*)-**2m**	61	98
6	(*S*)-**1n**, 96% ee	14.5	(*S*,*E*)-**2n**	67	98

[a] Reaction conditions: **1** (0.5 mmol), $[Cp^*RhCl_2]_2$ (0.025 mmol), $Cu(OAc)_2{\cdot}H_2O$ (1.25 mmol), and MeOH (25 μL) in CH_3CN (0.5 mL) at 50 °C under an O_2 atmosphere (balloon) unless otherwise noted. [b] Isolated yield.

The cross-coupling reaction of tertiary 2,3-allenol **1x** and secondary 2,3-allenol **1e** afforded the cross-cover product (*E*)-**2xe** in 49% yield together with the homo-coupling product (*E*)-**2e** in 17% yield (eqn (1)).



(1)

The structure of the target molecule was unambiguously determined by the X-ray single crystal diffraction analysis of (*E*)-**2k** (Fig. 1).

As we know, furan rings with chiral centres are very important in pharmaceutical chemistry.[8] With this method, (2,5-dihydrofuran-3-yl)but-2-en-1-one derivatives with a chiral centre could be easily constructed from the readily available optically active starting materials[9] without obvious loss of the enantiomeric purity (Table 4).

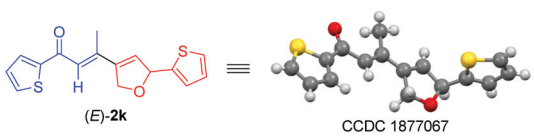
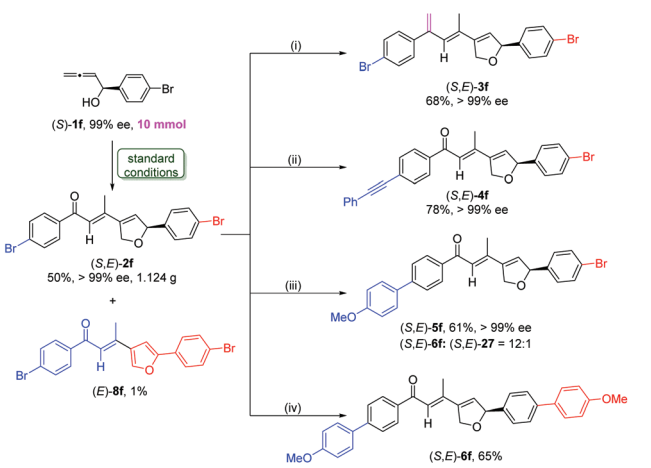


Fig. 1 ORTEP representation of (*E*)-**2k**.

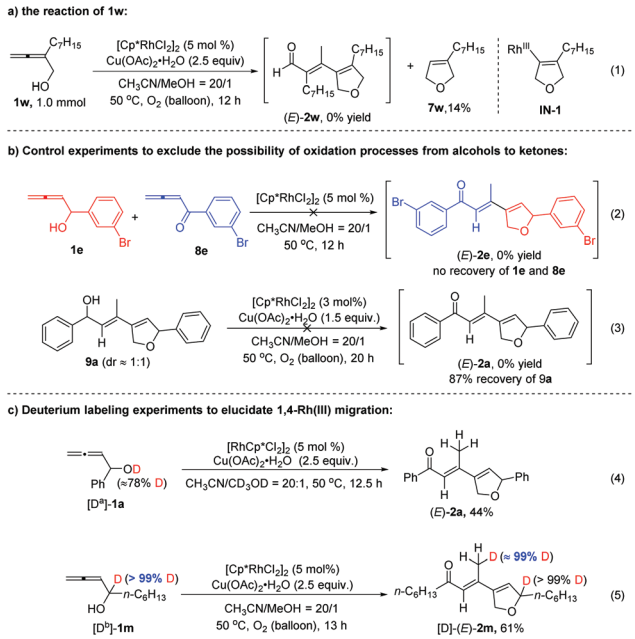
To further explore the synthetic utility of this procedure, the reaction was conducted on a 10 mmol scale, affording the product (*S*,*E*)-**2f** (>99% ee) in 50% yield under the standard conditions (Scheme 2). (*S*,*E*)-**2f** is synthetically versatile: the reaction of Cp_2TiMe_2 provided conjugated triene (*S*,*E*)-**3f** in 68% yield.[10] The two different C–Br bonds may be easily differentiated: the Sonogashira coupling of the C–Br bond on the electron-deficient phenyl ring with phenylacetylene gave (*S*,*E*)-**4f** in 78% yield exclusively. The Suzuki coupling of the same C–Br bond with (4-methoxyphenyl)boronic acid afforded the mono-coupling product (*S*,*E*)-**5f** with a selectivity of 12/1 over the double coupling product (*S*,*E*)-**6f**. When 3 equiv. of (4-methoxyphenyl)boronic acid were used, the double Suzuki coupling product (*S*,*E*)-**6f** was obtained in 65% yield exclusively.

In order to gain insight into the mechanism of this transformation, we conducted some control experiments (Scheme 3). When 2-heptyl-2,3-butadienol **1w** was applied under standard conditions, the expected dimeric product (*E*)-**2w** was not observed. Instead, 14% yield of the cycloisomerization product, 3-heptyl-2,5-dihydrofuran **7w**, was isolated, which may be formed *via* the protodemetalation of alkenylrhodium intermediate **IN-1** (Scheme 3, eqn (1)). The unsuccessful cross-coupling cyclization of allenol **1e** with its corresponding allenyl ketone **8e** excludes the formation of *in situ* generated allenyl ketone as the intermediate (Scheme 3, eqn (2)). When the corresponding alcohol **9a** was treated under standard conditions, (*E*)-**2a** was not formed with 87% recovery of **9a**, suggesting that (*E*)-**2a** was not formed *via* oxidation of **9a** (Scheme 3, eqn (3)). The reaction of [D[a]]-**1a** afforded the



Scheme 2 Synthesis of (*S*,*E*)-**2f** on a gram scale and derivatization reactions. Reaction conditions: All transformations were conducted under a nitrogen atmosphere, (i) Cp_2TiMe_2 (2.0 equiv., 0.1 M in toluene), 65 °C, 15 h; (ii) phenylacetylene (1.3 equiv.), $Pd(PPh_3)_2Cl_2$ (3 mol%), i-Pr_2NH (3 equiv.), CuI (6 mol%), THF, 40 °C, 15 h; (iii) (4-methoxyphenyl)boronic acid (0.91 equiv.), $Pd(PPh_3)_4$ (3 mol%), K_3PO_4 (3 equiv.), dioxane, 85 °C, 13 h; (iv) (4-methoxyphenyl)boronic acid (3 equiv.), $Pd(PPh_3)_4$ (5 mol%), K_3PO_4 (6 equiv.), dioxane, 85 °C, 15 h.

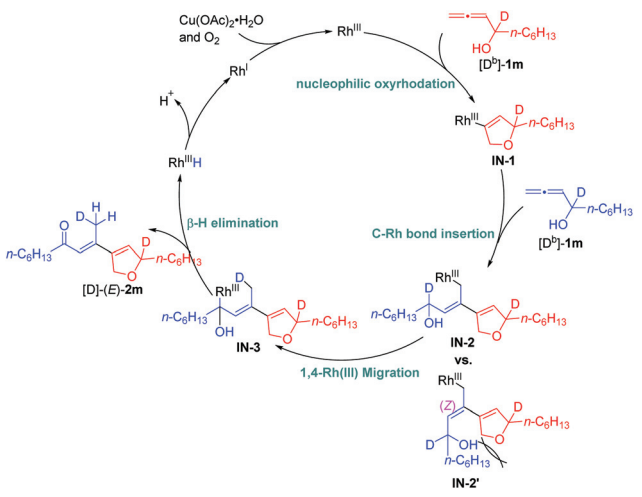
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Scheme 3 Mechanistic studies.

product (*E*)-**2a** without any deuterium incorporation (Scheme 3, eqn (4)). Unexpectedly, the reaction of [D^b]-**1m** afforded [D]-(*E*)-**2m** with a newly formed deuterated group with 99% of D incorporation (Scheme 3, eqn (5)).

Based on these experimental results, a mechanism was proposed as shown in Scheme 4 using [D^b]-**1m** as the substrate. Firstly, 2,3-allenol [D^b]-**1m** underwent a nucleophilic cyclic oxyrhodation to afford the 2,5-dihydrofuran-4-yl-rhodium intermediate **IN-1**, which underwent an insertion reaction with the second molecule of allenol to generate the stereodefined allylic rhodium intermediate **IN-2** stereoselectively by leaving the dihydrofuran ring far away from the bulky hydroxyalkyl group (**IN-2** *vs.* **IN-2′**). This intermediate would form **IN-3** *via* stereo-



Scheme 4 A possible mechanism.

defined 1,4-Rh(III) migration,[11,12] resulting in 1,4-D migration. β-H elimination yielded the final product [D]-(*E*)-**2m** with the generation of the Rh(III)H species. The Rh(III)H species underwent a reductive elimination process to generate a Rh(I) species, which was then oxidized to form the catalytically active Rh(III) species by $Cu(OAc)_2{\cdot}H_2O$ and O_2.[12a–c]

Conclusions

In summary, we have developed the first example of rhodium (III)-catalyzed highly regio- and stereo-selective oxidative dimerization of 2,3-allenols, providing an efficient approach for the synthesis of (2,5-dihydrofuran-3-yl)but-2-en-1-one derivatives. Such a reaction yields completely new products with a broad scope of substrates and good functionality tolerance. The stereoselectivity is determined by the steric effect between the *in situ* formed furan ring and the bulky hydroxyalkyl group. Optically active (2,5-dihydrofuran-3-yl)but-2-en-1-one derivatives are efficiently prepared from the readily available chiral 2,3-allenols. Further studies in this area are currently ongoing in our laboratory.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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